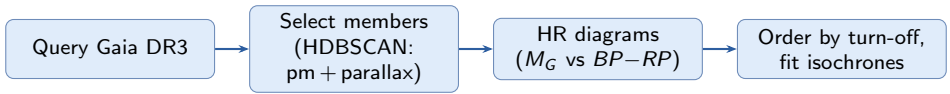
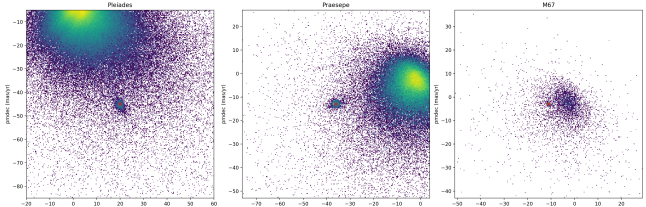
Which of three open clusters is oldest? Gaia DR3 alone can tell.

Data: `gaia_source_lite` (Gaia DR3); cone searches 10°/5°/1°; ARI Heidelberg TAP (ESA archive under maintenance). Filters: parallax/parallax_error > 5, RUWE < 1.4, non-null G, BP, RP.





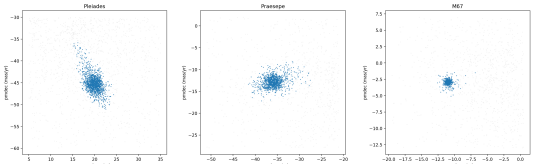
Clusters stand out as clumps in proper motion; red cross = expected centre.

Working method: plan first, one stage per prompt, run and check each stage, commit.

Answer: Order: Pleiades < Praesepe < M67. Praesepe 709 Myr (range 654–806 Myr); M67 4038 Myr (range 3766–4495 Myr); Pleiades age not constrained (flat loss curve, no turn-off).

Step 1: Clustering isolates each cluster at its expected distance.

Method: HDBSCAN on standardised (μ_{α^*}, μ_δ, parallax) after a proper-motion pre-selection window; `min_cluster_size=50`, `min_samples=10` (default settings returned only noise).



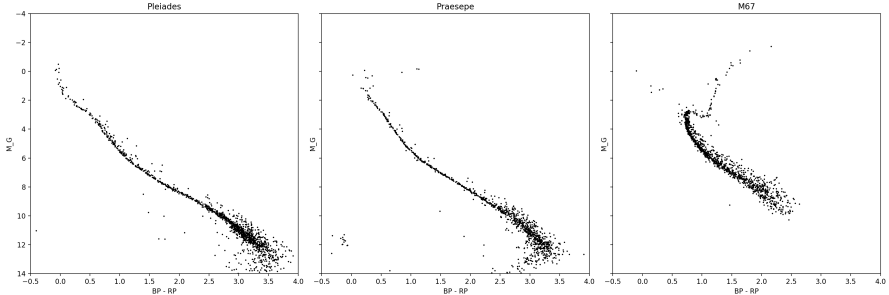
Cluster	N	d_{raw} (pc)	$d_{+0.017\,\mathrm{mas}}$ (pc)[a]	d_{ref} (pc)
Pleiades	1417	136.0	135.7	136.0
Praesepe	1136	185.0	184.5	185.0
M67	1315	866.6	854.0	850.0

[a] Approx. global zero-point of +0.017 mas. Reference distances from `src/config.py`. The reference distances are rounded literature values, so this is a consistency check and not an independent validation of the zero-point offset.

Takeaway: Distances agree with the references within 0.5% (largest: M67, +0.47%); counts 1417 / 1136 / 1315 are somewhat above the expected few hundred to about 1000.

Step 2: The HR diagrams already show a sequence of evolutionary stages.

Method: $M_G = G + 5\log_{10}(\varpi/100)$, ϖ in mas; colour $BP-RP$; cluster members only.



Pleiades: main sequence to blue end; no evolved giants.

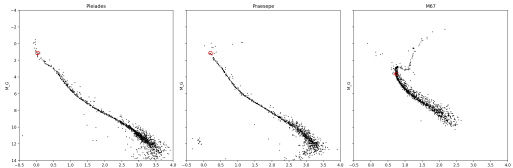
Praesepe: red-clump giants; white-dwarf sequence at faint blue end.

M67: turn-off, subgiant & red giant branches, red clump, blue stragglers.

Takeaway: Evolved stars in the order *none* (Pleiades), *few* (Praesepe), *many* (M67)—suggesting the age order before any measurement.

Step 3: The turn-off orders M67 clearly, the other two only weakly.

Method: Turn-off: bluest median $BP-RP$ in 0.25-mag bins of M_G ($\geq$5 stars, $-1 < M_G < 6$), 200 bootstrap resamples. Second measure: M_G of the fifth-brightest main-sequence star.



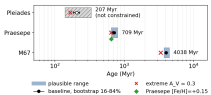
Measure	Pleiades < Praesepe	Praesepe < M67	Pleiades < M67	All three
Colour	0.95	0.875	0.95	0.83
Luminosity	0.685	1.0	1.0	0.685

Turn-off $BP-RP$: Pleiades 0.035, Praesepe 0.200, M67 0.740. No clear knee for Pleiades and Praesepe. Bootstrap reflects sampling noise only.

Takeaway: M67 is clearly oldest in luminosity (1.0 in both pairs; colour: 0.875 and 0.95). Pleiades < Praesepe: suggestive (colour 0.95, luminosity 0.685), not proven.

Step 4: Isochrone fits confirm the order, with stated limits.

Method: MIST v1.2, Gaia EDR3 passbands; fixed distance & A_V; age fitted to upper MS, turn-off and giants ($M_G < 6.0$); robust CMD loss; measured colour offset; $v/v_{\rm crit} = 0.4$.



Cluster	Age	Range (Myr)	Constrained
Pleiades	207 Myr	140–334	No[b]
Praesepe	709 Myr	654–806	Yes
M67	4038 Myr	3766–4495	Yes

[b] no: flat loss curve, no turn-off

Praesepe at [Fe/H]=+0.15: 654 Myr (approximate metallicity)

Limitations

- Colour offset 0.03–0.05 mag; reddening cannot explain it.
- Ranges are systematic envelopes.
- One model family (MIST, $v/v_{\rm crit} = 0.4$); no binary modelling.

Context only, not a result of this analysis: literature ages about 110, 650 and 4000 Myr (Pleiades, Praesepe, M67).

Takeaway: Order Pleiades < Praesepe < M67; plausible ranges do not overlap.